power_input

File: power_input.kicad_sch

boost_charger

File: boost_charger.kicad_sch

cap_bank

File: cap_bank.kicad_sch

pulse_switch

File: pulse_switch.kicad_sch

flyback

File: flyback.kicad_sch

sense_gates

File: sense_gates.kicad_sch

mcu_status

File: mcu_status.kicad_sch

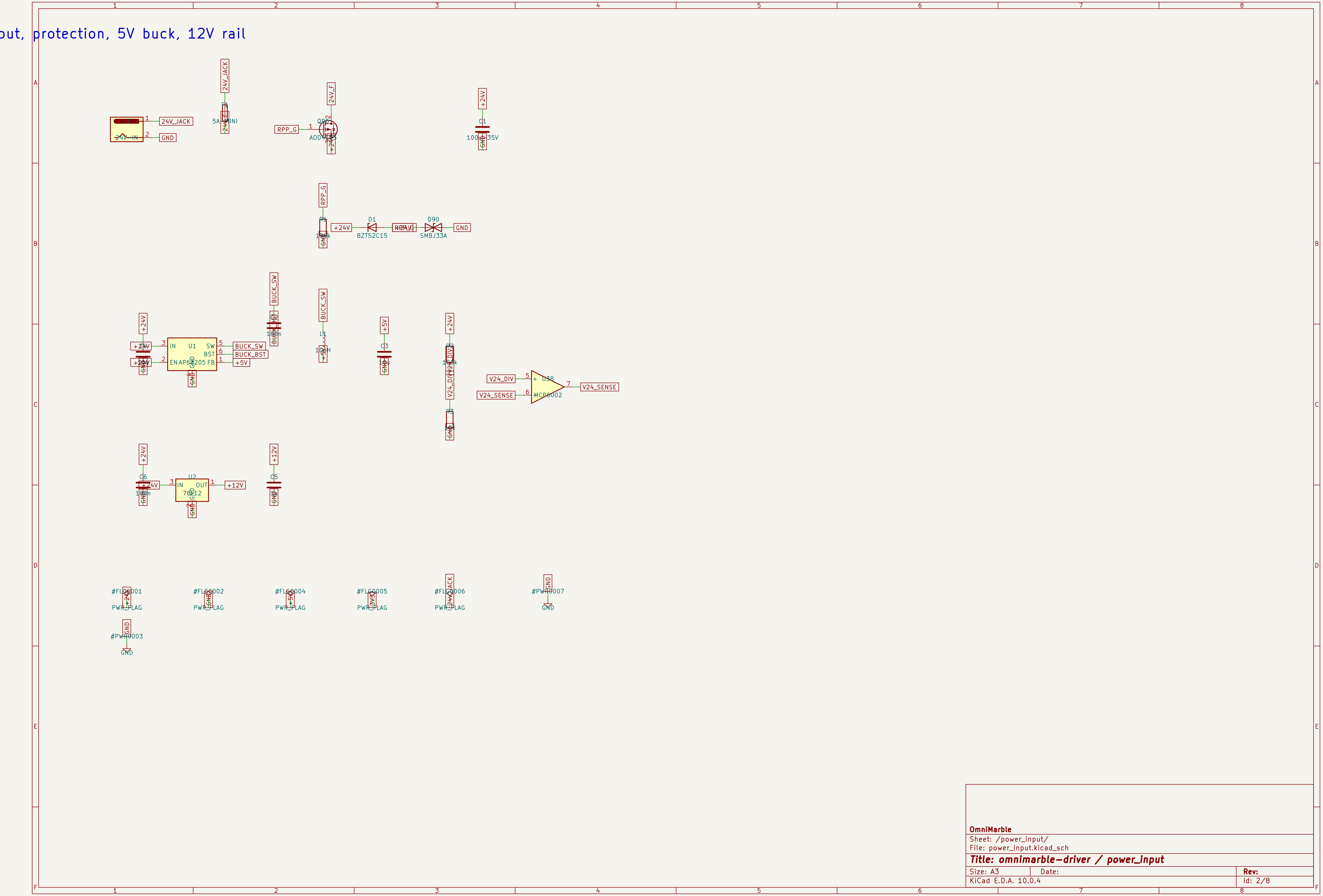
OmniMarble

Sheet: /

File: omnimarble-driver.kicad_sch

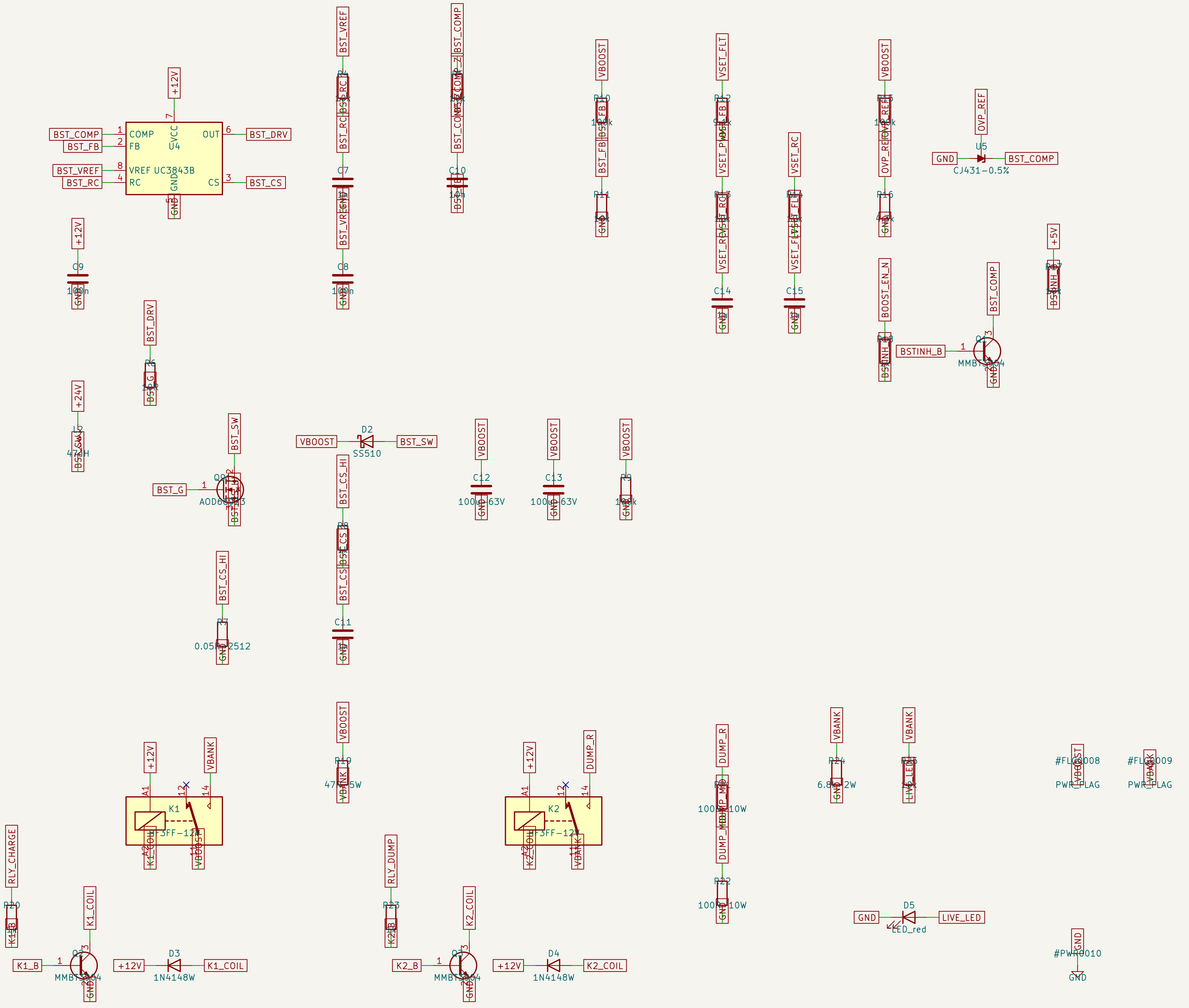
Title: OmniMarble 60V SELV coilgun driver

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 8/8



out, protection, 5V buck, 12V rail
24V_JACK
GND
24V_IN
5A_MINI
24V_F
RPP_G
+24V
100u 35V
BZT52C15
SMBJ33A
BUCK_SW
BUCK_BST
+5V
IN
SW
BST
EN
FB
U1
AP63205
V24_DIV
V24_SENSE
U3B
MCP6002
+12V
U2
78L12
OUT
100n
#FLG0001
#FLG0002
#FLG0004
#FLG0005
#FLG0006
PWR_FLAG
#PWR0003
#PWR0007
OmniMarble
Sheet: /power_input/
File: power_input.kicad_sch
Title: omnimarble-driver / power_input
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 2/8

4V->60V, OVP, charge/dump relays, bleed

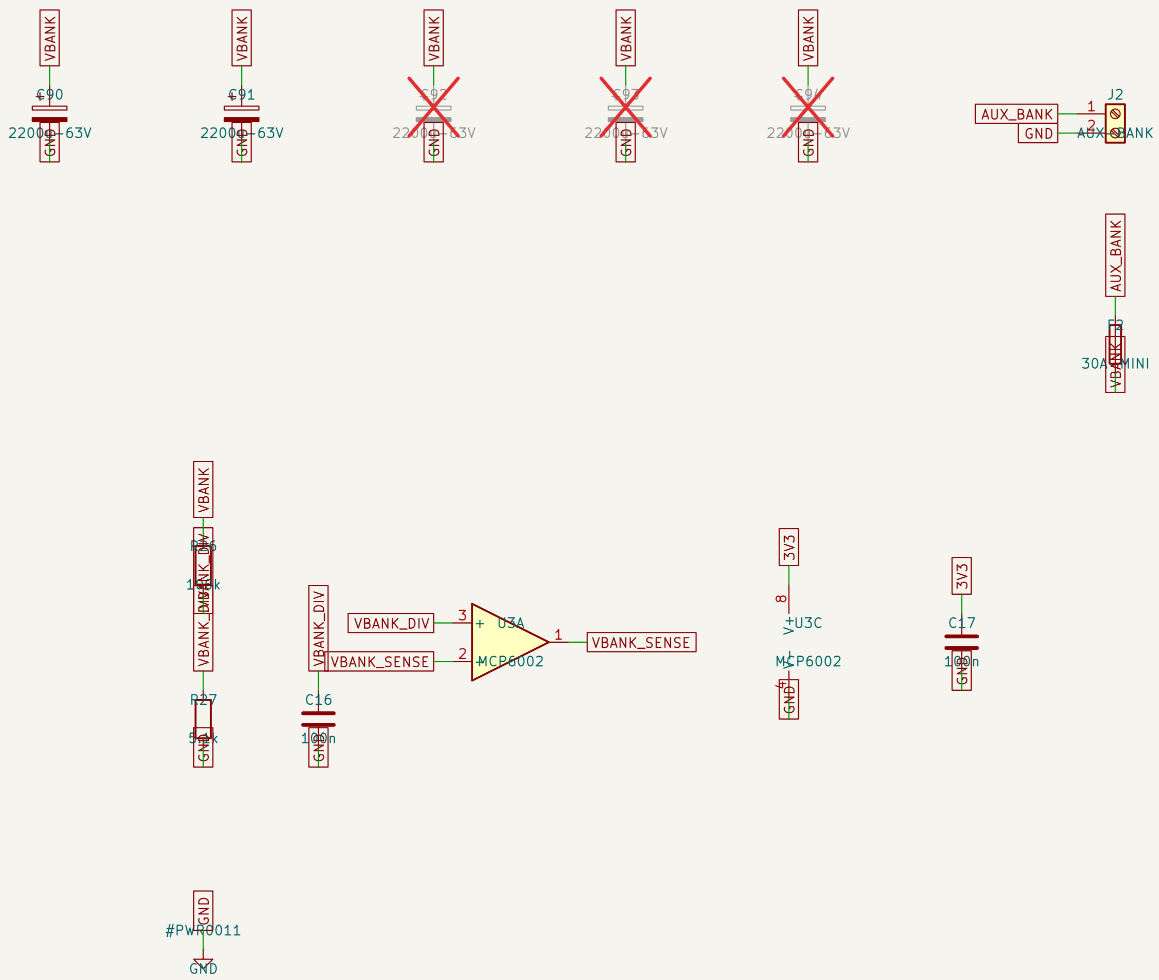


OmniMarble
Sheet: /boost_charger/
File: boost_charger.kicad_sch

Title: omnimarble-driver / boost_charger

Size: A3 Date: Rev:
KiCad E.D.A. 10.0.4 Id: 3/8

2200uF/63V snap-in (populate per bank table)

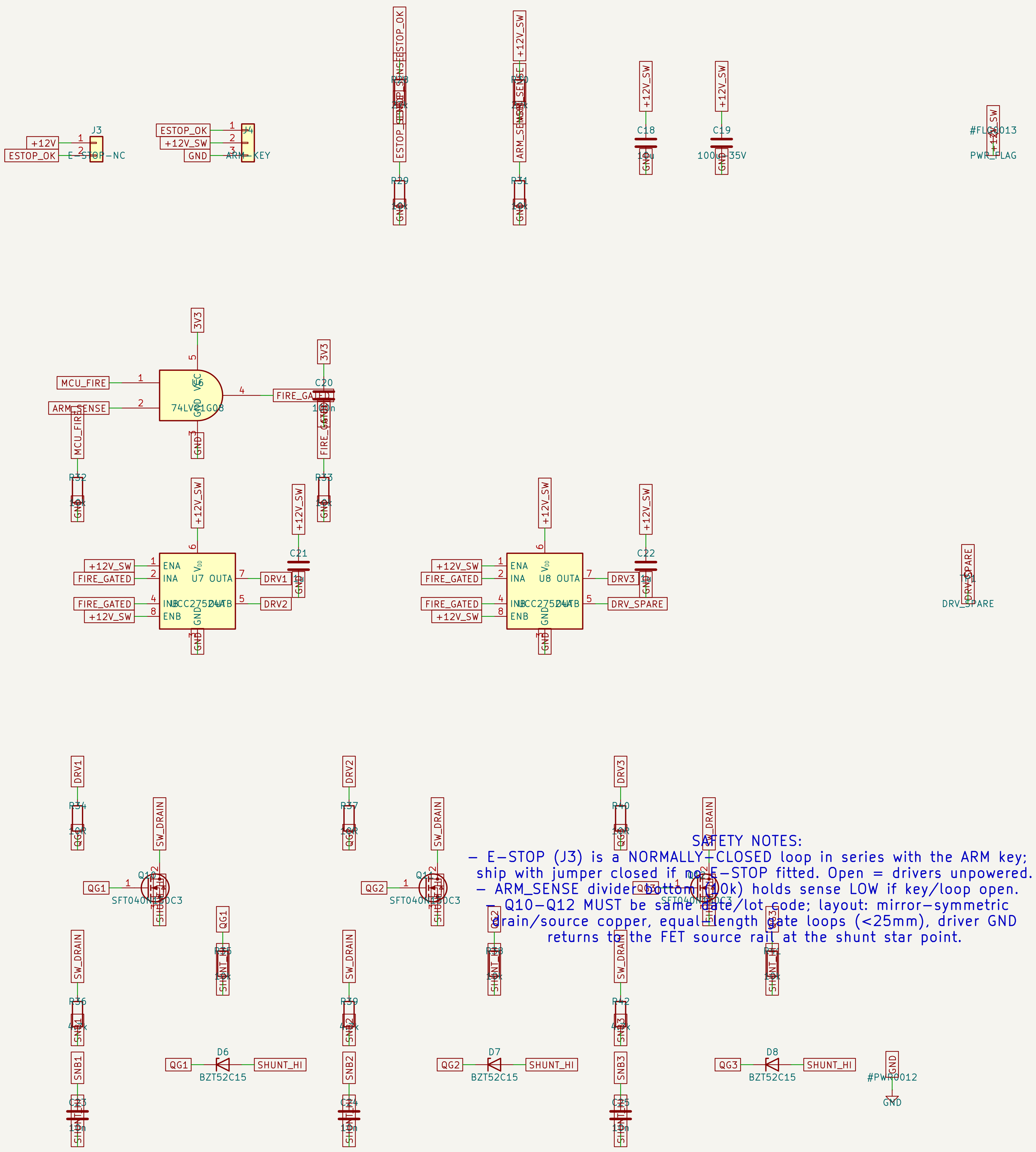


OmniMarble
Sheet: /cap_bank/
File: cap_bank.kicad_sch

Title: omnimarble-driver / cap_bank

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/8

T040N150C3, UCC27524A drive, ARM interlock



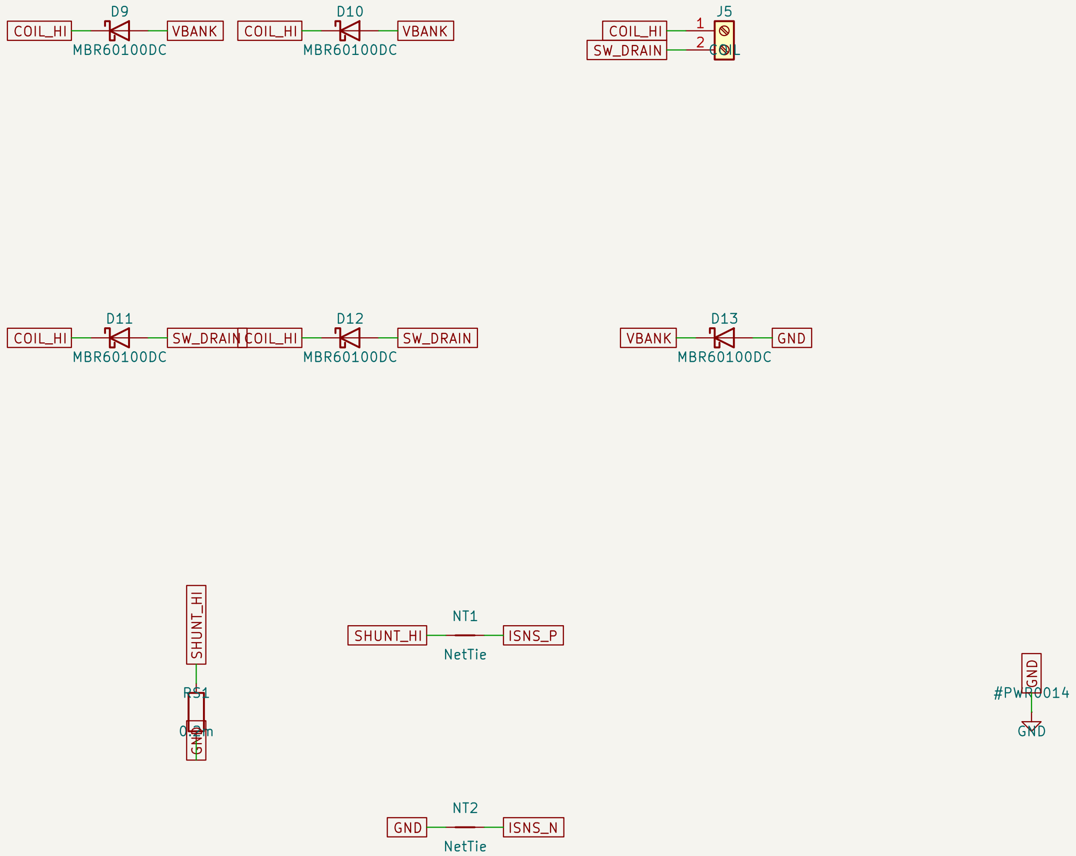
SAFETY NOTES:
- E-STOP (J3) is a NORMALLY-CLOSED loop in series with the ARM key; ship with jumper closed if no E-STOP fitted. Open = drivers unpowered.
- ARM_SENSE divider bottom (100k) holds sense LOW if key/loop open.
- Q10-Q12 MUST be same date/lot code; layout: mirror-symmetric drain/source copper, equal-length gate loops (<25mm), driver GND returns to the FET source rail at the shunt star point.

OmniMarble
Sheet: /pulse_switch/
File: pulse_switch.kicad_sch

Title: omnimarble-driver / pulse_switch

Size: A3 Date: Rev:
KiCad E.D.A. 10.0.4 Id: 5/8

blocking -> COIL -> switch -> shunt -> GND



OmniMarble

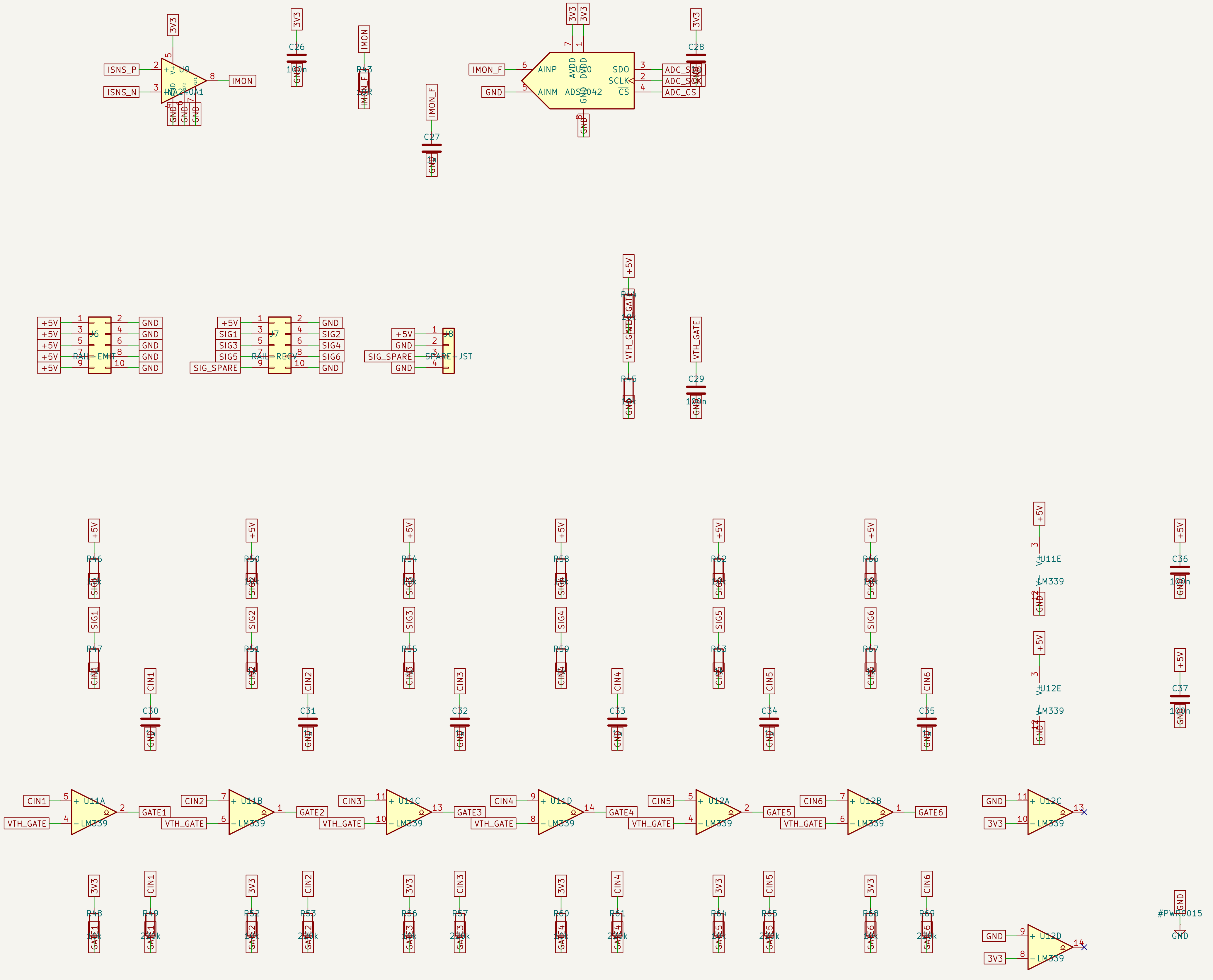
Sheet: /flyback/

File: flyback.kicad_sch

Title: omnimarble-driver / flyback

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/8

42 current capture; 6x IR gate comparators



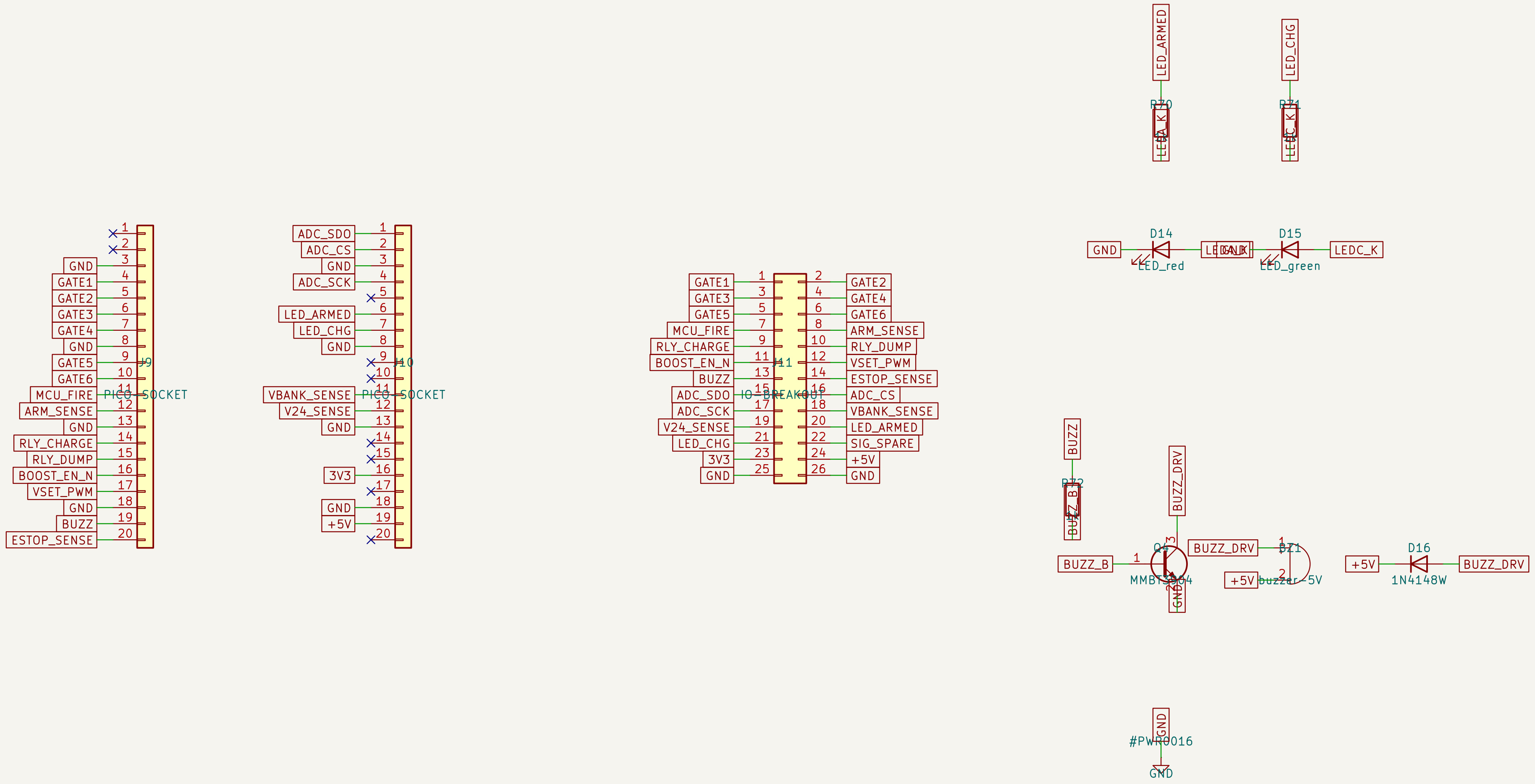
IR GATE TRUTH TABLE:
T -> phototransistor ON -> SIGx ~0V -> comparator OUT LOW
 -> phototransistor OFF -> SIGx ~5V -> comparator OUT HIGH
GH = marble in beam. Firmware timestamps the RISING edge.
pen-collector outputs pulled to 3V3: Pico-safe levels)

OmniMarble
Sheet: /sense_gates/
File: sense_gates.kicad_sch

Title: omnimarble-driver / sense_gates

Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 7/8

0), MCU–agnostic breakout, relay/status drive



OmniMarble
Sheet: /mcu_status/
File: mcu_status.kicad_sch
Title: omnimarble–driver / mcu_status
Size: A3 Date: Rev:
KiCad E.D.A. 10.0.4 Id: 8/8